

PUBLISHED

Question	Answer	Marks	Guidance
3(a)	$m = 69.522$	B1	
	$n = 0.156$ or 0.157	B1	Subtraction gives 0.156.
		2	
3(b)	H_0 : Po(0.7) fits the data. H_1 : Po(0.7) does not fit the data.	B1	Must mention distribution and data/number of breakdowns. May see H_0 : Po(0.7) is a satisfactory model for the data. H_1 : Po(0.7) is not a satisfactory model for the data.
	Combining last 3 columns give 13 and 6.828 or 6.829.	M1	Combine expected frequencies less than 5.
	$\frac{(88 - 99.317)^2}{99.317} + \frac{(73 - 69.552)^2}{69.552} + \dots + \frac{(13 - 6.829)^2}{6.829}$ $[1.2896 + 0.1740 + 0.1142 + 5.5764]$ OR $\frac{88^2}{99.317} + \frac{73^2}{69.552} + \frac{26^2}{24.333} + \frac{13^2}{6.829} - 200$	M1	At least 2 expressions (not including 200) of correct form seen. May see individual terms not summed. May be implied by 2 contributions correct to 3sf seen. 6.828 gives 5.5790.
	7.15 or 7.16	A1	SC B1 for 7.15 or 7.16 following M0 .
	'7.15' < 7.815, accept H_0 / do not reject H_0 / not significant.	M1	Compare <i>their</i> 7.15 with 7.815 (9.488 if 2 columns combined, 11.07 if no columns combined) and appropriate result (may be in terms of H_1). Test result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses.
	Insufficient evidence to reject expert's claim. OR Insufficient evidence to suggest that Po(0.7) is not a good fit/model [for the data].	A1	Correct conclusion from correct working ignoring hypotheses. In context. Level of uncertainty in language is used (for example, not 'prove') Allow 'Not enough evidence that / to show / conclude that...'. Do not accept statements such as 'there is sufficient evidence to suggest...' or 'no evidence...'.
		6	